



Captive-reared burrowing owls show higher site-affinity, survival, and reproductive performance when reintroduced using a soft-release

Aimee M. Mitchell^a, Troy I. Wellicome^{b,c,*}, Dawn Brodie^d, Kimberly M. Cheng^a

^a Department of Animal Science, University of British Columbia, 2357 Main Mall, Vancouver, BC, Canada V6T 1Z4

^b Canadian Wildlife Service, Environment Canada, Room 200, 4999-98th Ave., Edmonton, AB, Canada T6B 2X3

^c Department of Biological Sciences, University of Alberta, CW405, Biological Sciences Bldg., Edmonton, AB, Canada T6G 2E9

^d Burrowing Owl Conservation Society of BC, 111-365 East 1st St., North Vancouver, BC, Canada V7L 4W5

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ABSTRACT

A remaining challenge for animal reintroductions is how best to aid individuals transitioning from captivity into the wild. We tested two techniques – ‘soft’ vs. ‘hard’ release – in a 7-year study on endangered Western burrowing owls (*Athene cunicularia hypugaea*) in British Columbia, Canada. Traditionally, captive-reared yearling adults were released into the wild, as pairs, directly from artificial burrows (hard-release). Only 46% of 201 owls released in this way (2001–2004) stayed at release sites. To test for improved success, soft-releases were performed in the same manner as concurrent hard-releases (2005–2007), except above-ground enclosures confined each soft-release pair to the vicinity of their burrow for 2 weeks before complete release. Of 140 soft-released owls, 86% stayed at release sites; whereas, 66% of 100 hard-released owls stayed. Breeding-season survival was 70% for soft-released owls vs. 50% for hard-released owls. On average, soft-released owls produced 50% more fledglings than concurrently hard-released owls. Post-fledging survival was 69% and first-year return rate was 7.0% for fledglings from soft-released parents, compared to 50% and 4.0% for fledglings from hard-release parents. Ultimately, 0.17 offspring were recruited into the local wild breeding population per soft-released pair, compared to 0.05 local recruits per hard-released pair. We recommend that enclosure-based soft-releases be used for reintroduction of burrowing owls in British Columbia and elsewhere in North America. We encourage other tests for improved release success with soft-releases, particularly for species with a high tendency for dispersal or those likely to experience significant predation pressure.

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1. Introduction

A wide-variety of endangered or extirpated species have been successfully re-introduced into the wild (Fischer and Lindenmayer, 2000) – animals as diverse as swift fox (*Vulpes velox*; Ausband and Moehrenschlager, 2009), California condor (*Gymnogyps californianus*; Alagona, 2004), natterjack toad (*Bufo calamita*; Denton et al., 1997), duskytail darter (*Etheostoma percnurum*; Shute et al., 2005), and American burying beetle (*Nicrophorus americanus*; Amaral et al., 1997). Encouragingly, releases of captive stock for high-profile species such as peregrine falcon (*Falco peregrinus*) and Mauritius kestrel (*Falco punctatus Temminck*) have resulted in the re-establishment of wild populations close to what they were

prior to the species becoming endangered or extirpated (Tordoff and Redig, 2001; Nicoll et al., 2004).

Despite these and other notable reintroduction successes, there have been a greater number of species reintroduction failures (Dodd and Seigel, 1991; Fischer and Lindenmayer, 2000). Moreover, the failure rate of reintroductions apparently has not decreased over recent decades (Fischer and Lindenmayer, 2000). Progress in this field is hindered, in part, by a lack of post-release monitoring and experiments designed to gauge the relative performance of various release approaches (Short et al., 1992; Soderquist, 1994; Hardman and Moro, 2006). Without such assessments, the reasons that releases succeed or fail remain ambiguous, and animal conservation managers cannot base their decisions on direct evidence (Pullin et al., 2004; Armstrong and Seddon, 2008).

A topic that requires further investigation is how best to aid individuals during their transition from captivity into the wild. Pre-release confinement in a field enclosure (‘soft’-release) is a technique designed to ease this transition by allowing animals extra time to acclimate to the immediate environment of their release site. However, there are conflicting results in the literature

* Corresponding author at: Canadian Wildlife Service, Environment Canada, Room 200, 4999-98th Ave., Edmonton, AB, Canada T6B 2X3. Tel.: +1 780 951 8671.

E-mail addresses: athene.aimee@gmail.com (A.M. Mitchell), troy.wellicome@ec.gc.ca, tiw@ualberta.ca (T.I. Wellicome), dawnbrodie@telus.net (D. Brodie), kmctc@interchg.ubc.ca (K.M. Cheng).

about whether such soft-releases outperform 'hard'-releases, which do not use field enclosures (Hardman and Moro, 2006). Soft-releases have worked well for many species (e.g., Gatti, 1981; Bright and Morris, 1994; Ellis et al., 2000; Wanless et al., 2002), while others either have not benefited or occasionally have been hindered by field enclosures (Castro et al., 1994; Lovegrove, 1996; Moro, 2001; Clarke et al., 2002; Hardman and Moro, 2006).

The Western burrowing owl (*Athene cunicularia hypugaea*) has disappeared or declined in many parts of its former North American range (Wellicome and Holroyd, 2001), despite relatively low concomitant levels of habitat loss (Wellicome and Haug, 1995; McDonald et al., 2004; Commission for Environmental Cooperation, 2005). Releases have been conducted in several regions within this range: British Columbia (Dyer, 1991; Leupin and Low, 2001), Manitoba (De Smet, 1997), Saskatchewan (Poulin et al., 2006), and Minnesota (Martell et al., 2001). However, success of releases has varied considerably among projects and among years, and so too have the techniques employed. For example, the period that owls were temporarily confined in the field prior to release has varied from as little as a few hours to as much as a few weeks (Leupin and Low, 2001; Poulin et al., 2006). Interestingly, in one study the highest proportion of pairs remained together after release, and released pairs fledged the most offspring, in the same year that enclosures were left in place for the longest time (Poulin et al., 2006).

Unfortunately, after the mixed success of burrowing owl reintroductions in North America, all but one of the regions have halted releases, and the only currently active program is in British Columbia (BC). BC's wild burrowing owl population most likely had disappeared by 1980 (Leupin and Low, 2001), and the species is now listed as nationally and provincially endangered (COSEWIC, 2006). The very small population in BC now exists almost entirely because of annual releases of captive-reared owls, so there is a strong desire to optimize the effectiveness of these releases (Environment Canada, 2011).

The overall goal of the burrowing owl release project in BC is to re-establish a self-sustaining wild population (Environment Canada, 2011). This same long-term goal is shared by virtually all endangered species reintroduction programs (Fischer and Lindenmayer, 2000). However, determining if a self-sustaining population has been re-established often takes several decades (e.g., Millsap et al., 1998; Nelson et al., 2002; Moehrensclager and Moehrensclager, 2006). Over shorter time-frames, releases of animals have three main objectives: (1) settlement of animals into release areas, (2) survival of individuals following release, and (3) successful reproduction in the wild (Teixeira et al., 2007). Through careful field monitoring (2001–2007), we measured the success of the BC burrowing owl release project in relation to these three objectives for animal releases.

For over a decade, the typical release protocol for the BC program has been to place captive-reared yearling adult owls (burrowing owls are reproductively mature after 10 months-of-age) inside artificial burrows in the field, and then release these owls from their underground burrows directly into the wild (details in Leupin and Low, 2001). The main objective of our study was to use a controlled experiment to test whether these traditional hard-releases could be improved upon by enclosure-based soft-releases. We accomplished this by releasing captive-reared yearling adult owls from burrows that were surrounded for approximately 2 weeks by above-ground enclosures (soft-releases), while concurrently releasing similarly-reared yearling adult owls in the same manner except without enclosures over their burrows (hard-releases). As in traditional releases, one male and one female owl (hereafter, a 'pair') were chosen to be placed together inside each burrow. The experimental design of our investigation allows us to unambiguously attribute improved release success to temporary on-site confinement, as that is the only variable that differed between our two concurrent

treatment groups. Unfortunately, only 8% of 454 release studies reviewed by Seddon et al. (2007) used an experimental approach, which is the clearest way to evaluate the influence of a given factor in a species' reintroduction (Letty et al., 2007).

2. Methods

2.1. Study area

The study area was located in the Nicola Valley, in the Southern Interior Ecoprovince of British Columbia (BC Ministry of Forests, 1992). This valley is in the Thompson basin vegetation zone (Parish et al., 1996), between the cities of Merritt and Kamloops. There is a gradual transition from very hot and dry lower grasslands to cool and moist upper grasslands. Shrub-steppe communities (biogeoclimatic unit BGxh2), dominated by big sage (*Artemisia tridentata*) and bluebunch wheatgrass (*Pseudoroegneria spicata*), occur at the lowest elevations in this region (Grassland Conservation Council of BC, 2004).

Within the study area, owls were released from seven different 'release sites' (Table 1), situated in grassland fields, on properties owned and managed by a variety of private and public landholders. These sites were often grazed and many contained introduced, weedy plant species (Parish et al., 1996). The planting of forage and cereal crops, such as crested wheatgrass, alfalfa, and barley (Ministry of Agriculture, Food and Fisheries, 2002), has created a fragmented landscape (Leupin, 2004), with natural grassland areas remaining largely on sloped land where planting or mechanized alteration of vegetation is difficult. Release sites are spatially separated from one another, within a landscape matrix of farm fields, orchards, vineyards, and livestock pastures (see map in Leupin and Low, 2001).

Within each release site, several clusters of artificial underground burrows had been installed in previous years. Each cluster contained enough artificial burrows for an owl pair to occupy a nesting burrow and several nearby satellite burrows for roosting and food caching (Desmond and Savidge, 1999). Multiple pairs were released within each release site (Table 1), but no more than one pair was released in each burrow cluster and there were always more clusters than there were pairs released. 'Wild' burrowing owls, typically identified as released adults or young hatched in release sites in previous years, returned only to release sites.

2.2. Pre-release preparation

Owls at two breeding facilities in BC (Leupin et al., 2000) were raised in family groups, in separate pens, until owlets reached at least 4 weeks-of-age. After this time, all owls were allowed to

Table 1

Number of burrowing owl pairs released with two concurrent techniques, at each release site, over a 3-year period, in British Columbia.

Release site	2005		2006		2007	
	No. hard-released	No. soft-released	No. hard-released	No. soft-released	No. hard-released	No. soft-released
A	6	0 ^a	5	3 ^b	4	8
B	2	2	1	2	1	3
C			1	3		5
D	2	3	5	5	4	6
E	4	2	6	5 ^c	4	10
F	5	6				
G						7
Totals	19	13	18	18	13	39

^a Three soft-releases were started here, but cattle destroyed cages before pairs could be released.

^b Cattle present in this year, but cages were protected with electric fencing.

^c Owl pair in one cage was confirmed to be two males, and was therefore excluded from reproductive comparisons.

use a communal flyway. In late fall, juvenile owls (i.e., fledged owls that had hatched earlier in the same year) were captured inside breeding facilities, bled for DNA-based sexing, and separated into male and female communal pens for the winter. In spring when the previous year's hatchlings had reached 10 or 11 months-of-age, we placed them inside portable carriers (cat kennels) in pairs. Each randomly-chosen owl from one breeding facility was paired with a randomly-chosen owl of the opposite sex from the other breeding facility. That same day, each pair was transported to its release site and placed, by hand, into an artificial nest burrow. In all release years, pairs were placed in their burrows in the field on the same day of April (third weekend).

To allow individual identification, we banded each release owl with both a US Fish and Wildlife Service aluminium band and a coloured, alphanumeric, anodized-aluminium leg-band (Acraft Sign and Nameplate Co., Edmonton, Alberta, Canada). For more intensive tracking, in 2006 alone, we affixed harness-style radio-tags (Holohil Systems Limited, Carp, Ontario, Canada, Model BD-2) to 39 adults prior to their release. Including elastic attachment material, radio-tags weighed approximately 3 g, which is ~2% of average adult body weight. We distributed radio-tags approximately evenly between males ($n = 18$) and females ($n = 21$) and among the different release sites. All procedures in our study were approved by the Animal Care Committee of the University of British Columbia (Protocol # A05-0470).

2.3. Release techniques

2.3.1. Hard-release

As had been traditionally performed in releases in BC, each owl pair was placed inside nest chambers of artificial burrows, which had blocked outer entrances, and held there overnight (Leupin and Low, 1999). Owls were released, without above-ground enclosures, simply by unblocking their burrow entrances. Artificial burrows consisted of 2–3 m plastic tunnels attached to a nest bucket, all of which was buried underground (Fig. 1). Only the entrance to the tunnels and the top of an investigator-access chute, which each lead to the nest chamber, were visible above ground.

2.3.2. Soft-release

We installed predator-proof enclosures (2 models; $1.2 \times 1.2 \times 2.0$ m wooden frame and $1.2 \times 1.2 \times 1.2$ m PVC pipe frame, both covered with strong nylon mesh) over artificial burrows (Fig. 1) in the middle of April, almost 1 week before owls were placed in the burrows. Because some enclosures were destroyed by cattle in 2005, we used electric fencing, set back approximately 1 m from the enclosure, in fields that contained cattle at the time of release in 2006 and 2007. Electric fencing proved to be a safe and effective cattle deterrent. With enclosures already installed, we placed two owls inside each artificial burrow, in the same manner as in hard-releases. Burrow entrances were blocked for the same period of time as in hard-releases. In 2005 and 2006, we attempted to use both hard-release and soft-release techniques equally within each release site, ensuring that the two treatments were spatially interspersed (Table 1). Though release sites also were spatially interspersed in 2007, a greater proportion of pairs were soft-released than were hard-released.

We removed each enclosure approximately 2 weeks (mean $\pm$ SD = 15.1 ± 3.4 days) after pairs were placed in burrows. Removing the enclosures required about 15 min. We disturbed hard-released birds for an equal amount of time on the same days to keep disturbance consistent between treatments.

2.3.3. Supplemental feeding

From the day we placed a pair into their burrow to the day they failed or fledged young (fledging = 28–35 days post-hatch), we

supplemented the diets of each pair by leaving food (dead laboratory mice, *Mus musculus*, and domestic chicks, *Gallus gallus domesticus*) inside their release-burrow, every 3–4 days. We provided hard- and soft-release pairs with equal amounts of supplemental food, averaging 42.5 g (1.5 food items) per adult per day, which is 1.6 times the amount required by an adult burrowing owl for daily-existence metabolism in captivity (mean = 26 g; Marti, 1973). After fledging, we visited each nest every 4 days, leaving one food item (mean = 28.3 g/item) per adult or fledgling observed on that visit. This post-fledging supplemental feeding continued until owls could no longer be located anywhere (i.e., when they had dispersed from the study area or migrated).

2.4. Post-release monitoring

'Release' was the time that the block on the burrow entrance was removed for hard-released birds, and the time that the enclosure was removed for soft-released birds. We considered immediate post-release actions of adult owls to be either: (1) stayed or (2) vacated (i.e., moved permanently outside of release sites). Birds were recorded as 'staying' if they (a) were located alive at their release site (not necessarily at the same burrow) for 2 or more nest visits within the first 2 weeks of release (note that no owls vacated the release sites more than 2 weeks after being released), (b) occupied a release site other than the one from which they were released, or (c) were found dead at a release site at any time during the summer. Owls were considered to have 'vacated' release sites if they (a) were not observed at any release site in two visits within the first 2 weeks of release or (b) were found dead > 300 m outside of any release site.

In 2005 and 2006 only, we attempted to determine causes of mortality of adults and juveniles by the appearance of remains: (1) individually-plucked feathers with several crimped shafts indicated avian predation, (2) chewed feathers and remains in large pieces indicated mammalian predation, or (3) whole body with broken bones or head trauma (e.g., smashed beak) indicated collision with a vehicle.

We visually monitored all nest burrow entrances twice per week, and checked inside nest chambers at 10-day intervals, until clutch completion. We estimated the date of clutch initiation (day first egg was laid) by calculating backwards in time, assuming one egg had been laid every 1.5 days (Wellicome, 2005). We also recorded number of eggs and number of chicks when accessing nest chambers inside artificial burrows. To minimize disturbance of nestlings, we did not open nest chambers between hatching and when owlets were near fledging age (between 28 and 35 days post-hatch). At fledging time, we inspected the nest burrow and any satellite roost burrows (i.e., burrows habitually used by burrowing owls for loafing or caching prey; Environment Canada, 2011). We then captured all owlets by hand inside accessible artificial burrow chambers, and banded each one with a US Fish and Wildlife Service aluminium leg-band and a coloured alphanumeric anodized-aluminium leg-band (Acraft Sign and Nameplate Co., Edmonton, Alberta, Canada) for individual identification. At each nest, we used the number of owlets banded between 28 and 35 days-of-age as our estimate of the number of fledglings produced.

2.5. Tracking radio-tagged adults

In 2006, we began ground-surveying for radio signals from all tagged adults, 3 days after owl pairs were placed in release burrows. We attempted to locate these signals on foot, with a hand-held YAGI antenna (3-element) and Communications Specialist R-1000 receiver, and attempted to visually observe each adult twice per week at the release sites. Signals were located for both hard- and soft-released adults. We located the signals of soft-released

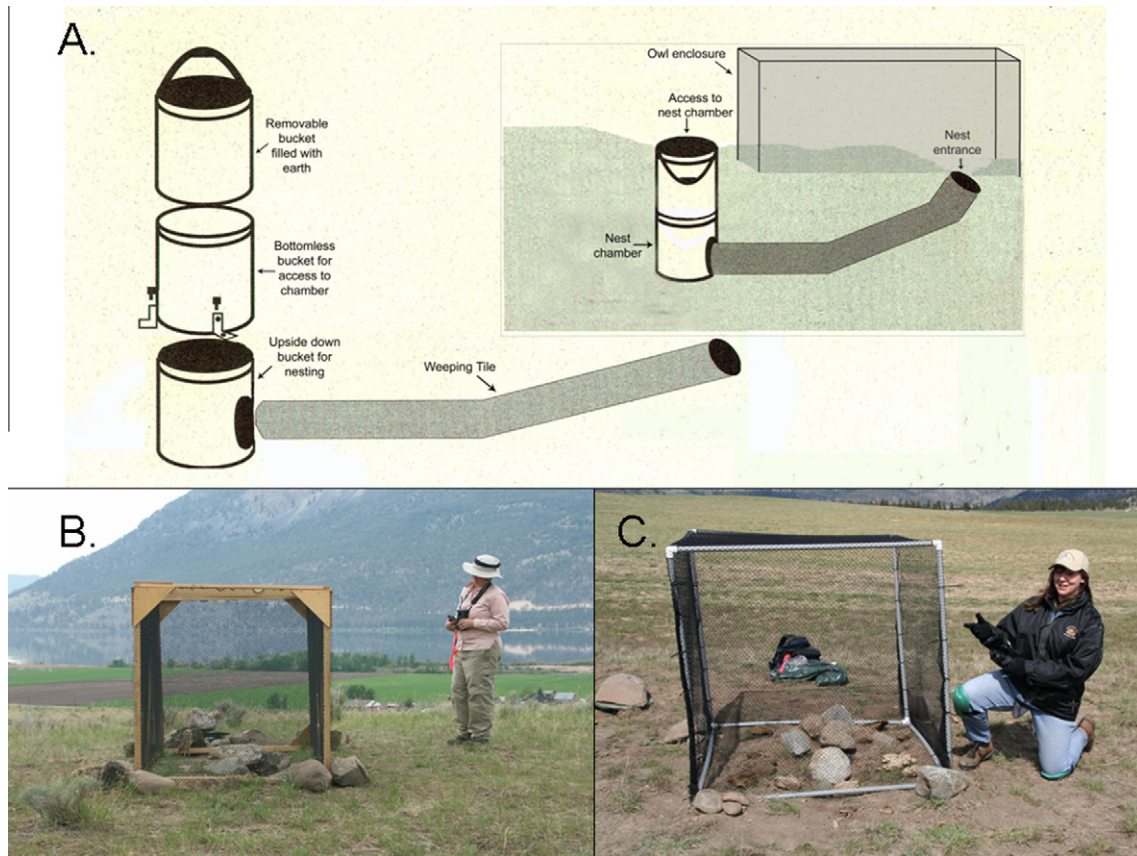


Fig. 1. Enclosures placed over entrances to artificial burrows in British Columbia. (A) Investigators accessed nest chambers via a removable bucket system located outside the enclosure. Only the entrance to the tunnels, and the top of an investigator-access chute that lead to the nest chamber, were visible above-ground. At some sites, electric fencing surrounded enclosures for protection from cattle disturbance. (B) Wooden enclosures were used in 2005 and 2006 (photo: Chris Currie). (C) PVC enclosures were used, in addition to wooden enclosures, in 2007 (photo: Mike MackIntosh).

owls after their enclosures were removed, but also before their enclosures were removed as a test for radio functioning.

For any signals not located using the above protocol, we conducted driving surveys, travelling further and further from the release sites, in a truck with one roof-top antenna (36" high Radial/Larsen, MAXRAD), at least once per week for the entire season (late-April to late-September). We conducted these vehicle surveys largely from highways, throughout all grassland regions (which occur in long narrow strips) in the valleys of southern BC. We also surveyed from secondary or gravel roads whenever those were accessible. All release sites, and virtually all potential owl habitat (open fields), were within 5 km of highways. The maximum distance owls could be detected from vehicles, based on tests of owls of known location, averaged 2 km.

To search for adult owls that were not located using the above ground-survey protocols, we also conducted two aerial surveys, the first of which began 1 week after all owls were placed in their release burrows. In the initial flight, we covered all grassland regions within the Nicola Valley and a 20-km radius around the city of Kamloops. While in flight, we determined the maximum distance radio signals could be detected (~10 km) by scanning for test radio-tags left in the Kamloops Airport and also for radio-tags on owls whose locations we already knew from ground surveys. One week after enclosures were removed from the burrows of soft-released owls, we performed the second aerial survey to search for all owls that had not been recently located using ground-search methods. In this second aerial survey, we expanded our search area to include grassland areas 20 km on either side of the Nicola Valley.

Using the above walking, driving, and flying surveys, we were able to monitor the locations of 33 of the 39 radio-tagged adult owls until they died or until the breeding season was over, with signals from the other six owls being permanently lost within the season (referred to as 'censored'). We tracked adults from the time of release, in late-April, until just prior to migration, in late-September. Of 18 owls that were located and alive near the end of the season, we were able to remove radio-tags from 17 by recapturing them in push-door traps (Winchell, 1999), set at their nest or satellite burrows, immediately prior to the predicted expiration of radio batteries (~16 weeks). Radio-tagging during the breeding season did not appear to affect an adult's ability to migrate and return the following year (Mitchell, 2008).

2.6. Monitoring juvenile survival

When owlets were being leg-banded in 2005 and 2006, we fitted a subset of them with collar-style radio-tags (Holohil Systems Limited, Carp, Ontario, Canada, Models PD-2, and RD-2). Of the 235 owlets that we banded, we also radio-tagged 27 of them (14 from hard-released and 13 from soft-released nests), each of which weighed ≥ 135 grams so fledglings would be carrying < 3% of their body weight. Our selection of an owlet for radio-tagging was random within each brood that produced at least two fledglings. Within each release site, we placed a radio-tag on one juvenile from at least one hard-released pair and on one juvenile from at least one soft-released pair.

On foot, we checked for signals from radio-tagged juveniles every 3–4 days throughout the post-fledging period. We considered

the post-fledging period to commence at 40 days post-hatch and end at permanent dispersal or migration (Todd et al., 2003), which typically occurs in BC between early-September and early-October (Leupin et al., 2001). We followed signals until (1) there was visual confirmation of a juvenile owl that was alive and wearing a radio-tag, (2) there was evidence of mortality of the tracked individual, or (3) the signal location and direction indicated that the juvenile was inside a burrow. If the signal in the latter case came from the same artificial burrow on two consecutive visits, we opened and inspected the burrow to determine if the juvenile inside was alive or dead. When a juvenile's radio-signal could not be located at its natal release site, we surveyed for the individual's signal at all other release sites in the study area. If the signal from a juvenile's radio-tag began to fail, we attempted to recapture the owl and replace its radio-tag. Of the 14 radio-tagged juveniles known to be alive at the end of the 2005 and 2006 seasons, combined, we successfully captured 11 of them, using baited push-door traps at burrow entrances (Winchell, 1999) or by hand inside artificial burrows (Wellicome et al., 1997), and removed their radio-tags prior to migration. When a juvenile could not be located at any of the release sites, we searched for its signal while conducting vehicle-based surveys for adult owls. We determined that 2.0 km was the maximum distance from which signals could be detected while on foot or in a vehicle. No aerial surveys were done during the post-fledging period.

2.7. Data analysis

We used log-likelihood ratios (G^2) to test for between-treatment differences in (1) proportions of all released owls (all of which had coloured leg-bands) that survived the breeding season (as determined through visual re-sightings), (2) proportions of all released owls that stayed at release sites vs. vacating them, and (3) proportions of all released pairs that produced at least one fledgling. We also used log-likelihood ratios to test for between-treatment differences in the proportions of radio-tagged adult owls that stayed at release sites instead of vacating, and also for differences in the proportions of radio-tagged owls (adults and juveniles, separately) that were known to survive the breeding and/or post-fledging season. To compare the number of fledglings produced per soft-released pair to that per hard-released pair, we used a Wilcoxon rank-sum test with continuity correction (Mann and Whitney, 1947) because the data were non-normally distributed.

As the objectives in this study were to test for improved performance for owls from soft-releases compared to hard-releases (i.e., results in only one direction), we used one-tailed test statistics (Zar, 1974). If our data supported the hypothesis that owls associated with soft-releases show higher site-affinity, reproductive performance, or survival compared to owls with hard-releases, then our recommendation would be to adopt this new release technique in burrowing owl reintroduction programs. In contrast, if the null hypothesis (i.e., owl performance with soft-releases $\leq$ owl performance with hard-releases) was not rejected, we would recommend against using the new technique, as it involves more effort and materials to include a burrow enclosure in a release. We used SAS 9.1.3, JMP 5.1.2, and R 2.11.0 statistical packages, with a significance level (α) of 0.05 for all tests.

3. Results

3.1. Traditional hard-releases

From 2001 to 2004, 201 yearling owls were hard-released. Of these 201 owls, 46% stayed at sites following release (Fig. 2). With the exception of less than 10 re-sightings, breeding-season survival of vacating owls (i.e., those that permanently moved outside of release sites) was unknown during this time period, but none remained or bred within any of the release sites.

Between 2001 and 2004, the mean number of fledglings produced per released adult was 0.9, with 40% of the 201 released adults producing one or more fledgling. However, owls that stayed at release sites had relatively high nesting success, as 74% of those owls fledged one or more owlet.

During annual spring arrivals in 2002–2005, a cumulative total of 35 owls returned to annual release sites prior to the current year's release of captive birds. Of these 35 owls, 25 were of known identity, with wild-hatched owls (i.e., hatched in the field at a release site in a previous year) contributing 52% of these known returns ($n = 13$) and owls previously released as adults contributing 48% ($n = 12$). Five owls were unbanded (origin unknown) and the remaining five could not be identified. From 2003 to 2005, mean over-winter return rates were similar ($G^2 = 0.2$, $P = 0.65$) for wild-hatched juveniles (7.1%) and released adults (6.0%). During this period, there were 0.06 offspring recruited (as first-year breeders) into the local population per released adult owl.

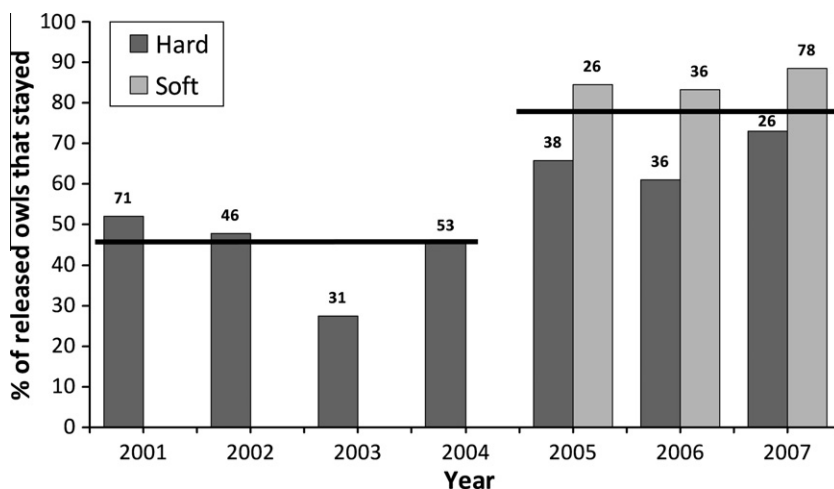


Fig. 2. Percentage of captive-reared burrowing owls that stayed at release sites following release into the wild, in British Columbia, 2001–2007. From 2001 to 2004, only hard-releases occurred, and the horizontal line through those bars indicates the overall 4-year percentage. From 2005 to 2007, the horizontal line through the bars indicates the overall percentage for all birds released (including both treatments), over that 3-year period. Sample size is shown above each bar.

Table 2

Post-release observations of captive-reared burrowing owls in British Columbia (2005–2007).

Post-release observations	Hard-released	Soft-released
Individual owls	<i>n</i> = 100	<i>n</i> = 140
Stayed at release sites	66% (66)	86% (121)
Survived breeding season	50% (50)	70% (98)
Owl pairs ^a	<i>n</i> = 50	<i>n</i> = 69
Initiated clutch	46% (23)	86% (59)
Hatched chicks	36% (18)	62% (43)
Fledged young	32% (16)	48% (33)

^a Owl pair in one cage was confirmed to be two males, and was therefore excluded from reproductive comparisons.

3.2. Soft- vs. hard-releases

3.2.1. Adult survival during the breeding season

From 2005 to 2007, 78% of all released birds (*n* = 240) stayed after their release (Fig. 2), with 20% more soft-released birds than hard-released birds staying at sites ($G^2 = 14.0$, $P < 0.01$; Table 2). Of all released owls (vacated + stayed), 20% more soft-released owls than hard-released owls survived the breeding season ($G^2 = 9.9$, $P < 0.01$; Table 2). In 2005 and 2006, leg-bands aided the confirmation of 28 deaths of released adult owls; 43% of these deaths occurred within the first week and 71% occurred within the first 4 weeks of release. Avian predation caused 79% of deaths, and collision with vehicles accounted for the remaining 21%. Causes of death differed between release techniques ($G^2 = 3.5$, $P = 0.03$), with a higher proportion of all deaths of hard-released owls being attributed to avian predators (17 of 20; 85%) than were for soft-released owls (4 of 8; 50%).

3.2.2. Monitoring adults with radio-tags

Telemetry work, performed in 2006, showed that a much higher percentage of adults stayed at release sites after being soft-released (10 of 11 tagged owls stayed; 91%) than stayed after being hard-released (11 of 28 tagged owls stayed; 39%) ($G^2 = 9.6$, $P < 0.01$). Ten of 18 tagged male owls stayed after release (56%), and 11 of 21 tagged female owls stayed after release (52%), indicating that sex did not influence the percentage of owls that stayed at release sites ($G^2 = 0.1$, $P = 0.83$).

During monitoring of radio-tagged adults, signals from six adult owls were permanently lost (i.e., censored). Five of these six censored owls were hard-released adults whose signals were lost within the first week after release, and the one other censored individual was a soft-released adult whose transmitter fell off more than 2 weeks after release. Of the total 33 owls of known fate (i.e., excluding censored owls), 61% stayed at the release sites and 39% vacated the release sites. No significant difference was detected ($G^2 = 0.04$, $P = 0.84$) between the percent survival of all owls that stayed at release sites (13 owls survived of the 20 that stayed = 65%) compared to all owls that vacated release sites (five owls survived of the 13 that vacated = 38%). It appeared that a higher percentage of radio-tagged owls survived after soft-release (seven owls survived of the 11 soft-released = 64%) than after hard-release (10 owls survived of the 28 hard-released = 36%), but this apparent difference was not statistically significant ($G^2 = 2.5$, $P = 0.11$).

3.2.3. Nesting activities

From 2005 to 2007, we monitored 50 pairs of owls that were given the traditional hard-release and 69 pairs that had an enclosure-based soft-release (Table 2). Soft-released pairs initiated clutches in 40% more cases, hatched young in 26% more cases, then ultimately fledged one or more young in 16% more cases ($G^2 = 3.0$,

Table 3

Productivity per pair of captive-reared burrowing owls released in British Columbia with either of two alternative release techniques (2005–2007).

Reproductive measure (per pair)	Hard-released (<i>n</i> = 50)		Soft-released (<i>n</i> = 69)	
	Mean	SE	Mean	SE
Eggs laid	3.2	0.6	6.5	0.4
Chicks hatched	2.1	0.5	3.9	0.4
Young fledged	1.6	0.4	2.4	0.3

$P = 0.04$) than hard-released pairs. This was reflected in the differences in reproductive output (Table 3), as soft-released pairs averaged more eggs, more hatchlings, and ultimately 0.8 more fledglings/pair than did hard-released pairs ($W = 1998$; $P < 0.05$). Examining only those owl pairs that initiated clutches, nest survival probabilities did not differ between release techniques (0.54 ± 0.03 and 0.59 ± 0.03 for soft- and hard-released pairs, respectively; 2005 and 2006 combined). In the end, the 69 soft-released pairs produced 168 fledglings, while the 50 hard-released pairs produced only 82, for a mean of 1.2 and 0.8 fledglings produced per soft- and hard-released adult, respectively (fledglings/pair shown in Table 3).

3.2.4. Post-fledging survival of offspring

Radio-tracking of juveniles during the post-fledging period suggested that juveniles of known fate (i.e., excluding censored owls) from parents in soft-releases survived in similar proportions (9 of 13 = 69%) to those of known fate from hard-releases (5 of 10 = 50%; $G^2 = 0.8$, $P = 0.38$). Causes of death for juveniles did not appear to be influenced by the release origin of their parents (Table 4), though sample sizes were too small to provide adequate power for statistical testing.

3.2.5. Annual returns

One of 62 (2%) soft-released adults and 2 of 74 (3%) hard-released adults from 2005 or 2006 returned in 2006 or 2007, so adult return rates did not differ for the two release techniques ($G^2 = 0.2$, $P = 0.66$). In contrast to released adults, 28% of 'wild' adults (i.e., those adults present at release sites in late-winter or early-spring, prior to the annual release of captive-bred owls) returned to the release sites again in the year subsequent (3 of 10 in 2006, 4 of 15 in 2007). Though the observed percentage of returning fledglings that originated from soft-released parents (5 of 71 fledglings returned, or 7.0%) was 75% higher than the percentage of returning fledglings that came from hard-released parents (2 of 50 fledglings, or 4.0%), this increase was not statistically significant ($G^2 = 0.5$, $P = 0.47$). In the end, the local recruitment of first-year offspring into the breeding population per soft-released pair was 0.17 (five recruited offspring from 30 pairs) vs. 0.05 local recruits/pair for hard-released pairs (two recruited offspring from 37 pairs).

Table 4

Radio-tagged juvenile burrowing owls were monitored in British Columbia, in 2005 and 2006, to assess their fate (survived, died, or unknown) over the post-fledging period (40 days post-hatch until migration). Causes of death are separated into predation, by a raptor or mammal, or collision with a vehicle. Values indicated under each fate-category are proportions of all radio-tagged juveniles in that year. Values in parentheses are numbers of radio-tagged juveniles from hard-released (*n* = 13) and soft-released (*n* = 14) parents, respectively.

Year	Survived to migrate	Predation		Collision with vehicle	Unknown fate
		Raptor	Mammal		
2005	0.47 (3:3)	0.15 (0:2)	0.08 (1:0)	0.15 (2:0)	0.15 (2:0)
2006	0.58 (2:6)	0.00 (0:0)	0.14 (1:1)	0.14 (1:1)	0.14 (1:1)
Totals	0.52 (5:9)	0.07 (0:2)	0.11 (2:1)	0.15 (3:1)	0.15 (3:1)

4. Discussion

4.1. Site-affinity

Site-affinity in mammals is typically higher following soft-release than following hard-release (Davis, 1983; Bright and Morris, 1994; Parker et al., 2008; but see Hardman and Moro, 2006). However, the influence of release technique on site-affinity is rarely studied in birds (Franzreb, 2004).

In our experiment, both visual monitoring and radio-tracking showed that soft-released burrowing owls were more likely to stay in release areas than were hard-released owls. Though concurrent soft- and hard-releases of burrowing owls had not been conducted prior to our study, *post hoc* assessment of releases in Saskatchewan (Poulin et al., 2006) and Manitoba (De Smet, 1997) corroborate that more pairs stayed together in years that owls were confined in release cages for longer periods of time prior to release. However, we are unsure why burrowing owls are more likely to vacate sites after hard-release than after soft-release.

It has been suggested that other animals disperse (vacate) following hard-release because they are disoriented and overwhelmed by the new environment in which they suddenly find themselves (Biggins et al., 1999; Moehrensclager and MacDonald, 2003; Tweed et al., 2003; Letty et al., 2007). Time spent on-site in pre-release pens likely reduced this potential problem for soft-released burrowing owls by allowing them to acclimate to their new local environment. In support of this idea, we observed some owls repeatedly flying into the soft mesh of their enclosure, apparently attempting to immediately leave the release site; however, this behaviour subsided after a brief period of confinement in the field (AMM, pers. observ.).

Tweed et al. (2003) suggested that mate availability for reintroduced Hawaiian thrushes (*Myadestes palmeri*) influenced fidelity to territories after release. Each burrowing owl in our soft-releases had 2 weeks inside its pre-release enclosure to form a pair bond with a conspecific of the opposite sex. The presence of this 'captive audience' may have contributed to the higher site-affinity observed in soft-released vs. hard-released owls.

4.2. Reproductive output

Release methods that promote breeding soon after release into the wild increase the probability of re-establishing a self-perpetuating population (Bright and Morris, 1994). Given this, it is surprising that no previous studies have been able to experimentally test the influence of pre-release confinement on reproductive output in released animals.

We demonstrated that released burrowing owls are clearly capable of breeding successfully, even in the same season they are released, and that soft-releases increase their fledgling production. Much of the reproductive improvement with soft-releases likely results from increased site-affinity, given that owls in our study bred only when they remained at release sites and not when they vacated. However, this cannot completely explain the observed improvement because, examining only those owls that stayed at release sites, a higher proportion of soft-released owls than hard-released owls initiated breeding. To illustrate, 86% of soft-released owls stayed at their release sites and all 86% initiated clutches; whereas, 66% of hard-released owls stayed at their release sites but only 46% initiated clutches (Table 2). Interestingly, once pairs from each type of release had initiated breeding, they were equally likely to produce one or more fledglings.

Both fledging success (% of pairs producing ≥ 1 fledgling) and fledglings produced per soft-released adult were higher in our BC study than for soft-releases in Saskatchewan (Poulin et al., 2006).

However, reproductive output in Saskatchewan appeared to be higher in years when the period of pre-release confinement was increased. Furthermore, when only hard-releases were employed in an earlier BC reintroduction program (Leupin and Low, 2001), reproductive output was much lower than that observed in our soft-releases and also lower than that in our hard-releases. We propose that the differences in reproductive output between these two BC studies resulted from differences in pre-release confinement and, perhaps, from two additional factors: (1) a higher frequency of releasing single birds in the earlier project (Leupin and Low, 2001; see Section 4.1 for potential benefit of releasing pairs) and (2) more supplemental feeding in our release (every 3–4 days vs. weekly or biweekly). Wellicome (2000) showed that supplemental feeding had a strong influence on fledgling production in wild burrowing owls, so we suspect this factor has an equal or stronger influence on captive-bred owls, which are released into the wild with relatively little foraging experience.

4.3. Survival

Radio-tracking, conducted on a subset of released owls only in 2006, failed to detect significantly higher survival for soft-released owls, but effect size (i.e., observed difference) was in the predicted direction, and the *P*-value was low (0.11) despite a small number of radio-tagged owls. More importantly, visual monitoring of banded individuals showed that soft-released burrowing owls had significantly higher breeding-season survival than did hard-released owls, over three study years.

A variety of species show higher survival after soft-release than after hard-release: dormouse (*Muscardinus avelinarius*; Bright and Morris, 1994), swift fox (*Carbyn et al., 1994*), Attwater's prairie chicken (*Tympanuchus cupido attwateri*; Lockwood et al., 2005), hedgehog (*Erinaceus europaeus*; Molony et al., 2006), and Florida Key deer (*Odocoileus virginianus clavium*; Parker et al., 2008). On the other hand, several species show no difference in survival for hard-vs. soft-release: North Island saddlebacks (*Philesturnus carunculatus*; Lovegrove, 1996), Eastern grey kangaroos (*Macropus giganteus*; Campbell and Croft, 2001), short-tailed mice (*Leggadina lakedownensis*; Moro, 2001), and hare wallabies (*Lagorchestes hirsutus* and *L. fasciatus*; Hardman and Moro, 2006). Notably, all species that fail to show increased survival with soft-releases have been studied in situations with very low or zero predation pressure. We therefore suggest that predation pressure in release environments may often explain among-species differences in the influence of soft-release on survival.

Avian predation is typically the most common cause of death for adult and juvenile burrowing owls in Canada (Clayton and Schmutz, 1999; Todd et al., 2003; Shyry, 2005; Poulin et al., 2006). Predator populations, such as Swainson's hawk (*Buteo swainsonii*), red-tailed hawk (*Buteo jamaicensis*), and great horned owl (*Bubo virginianus*), have increased concurrently with increases in trees and fences, which provide nesting and perching structures for these large predators (Wellicome and Haug, 1995; Clayton and Schmutz, 1999). Leupin and Low (2001) reported that burrowing owls released in BC suffered heavy predation from Northern harriers (*Circus cyaneus*), great horned owls, and red-tailed hawks, thus hampering reintroduction efforts (Leupin, 2004). In our study, avian predation also caused the majority of deaths in adults and many deaths in juveniles (Table 4), and avian predation explained a lower proportion of adult deaths for soft-released than for hard-released owls. Soft-releases therefore provide a valuable option for reducing mortality from predators, given that directly controlling predator populations is not typically a palatable management option.

During the post-fledging period, the survival of wild-hatched offspring did not differ according to the release method of

their parents. Thus, increased reproductive output associated with soft-releases was not countered by decreased juvenile survival following that type of release. Poulin et al. (2006) also showed that survival of offspring from released parents was similar to survival of offspring from wild parents during the post-fledging period in Saskatchewan, suggesting that offspring are not negatively influenced by the captive upbringing of their parents.

4.4. Annual returns

Hard- and soft-released adults in our study had similar annual return rates, and so did their wild-hatched offspring. However, these return rates were higher than those reported for released burrowing owls in any other population; annual return rates for released adults and their offspring in Saskatchewan and Manitoba, and in earlier BC releases, were 0–1.6% (Poulin et al., 2006; De Smet, 1997; Leupin and Low, 2001), and the annual return rate for translocated fledglings that were soft-released in Minnesota (Martell et al., 2001) was 0%.

We are uncertain why return rates were higher in our release program than in other burrowing owl release programs in North America. It is possible that annual site-fidelity to release areas was higher because of limited options for owls to breed elsewhere in BC given the small area of grassland, low availability of natural burrows, and lack of wild conspecifics. Alternatively, increased site-fidelity may have resulted because owls released in our study were all hatched in captivity within BC, whereas owls studied in Saskatchewan, Manitoba, and Minnesota all originated from a different province or state than the one in which they were released. Poulin et al. (2006) suggested that the distant origins of their captive-reared owls may have affected the birds' ability or willingness to navigate successfully back to release sites.

Annual return rates for released adults were much lower than return rates for 'wild' owls in our study population (Section 3.2.5), and were also lower than the annual return rate for wild owls in both Saskatchewan (Poulin et al., 2006) and Manitoba (De Smet, 1997). In contrast, returns of wild-hatched offspring produced by released adults in our study were very similar to returns observed for offspring of wild owls in both Saskatchewan (Poulin et al., 2006) and Manitoba (De Smet, 1997). Because released adults are much less capable of remaining in the local population than wild adults, but the offspring of released owls are equally as likely as offspring from wild owls to recruit into the subsequent breeding population, the main contribution of released adults toward the re-establishment of a wild population occurs through their relatively high probability of producing offspring in the same year they are released.

From 2005 to 2007, higher site-affinity, survival, and reproductive output, in combination with similar post-fledging survival and return rates of offspring, cumulated to an overall local first-year breeding recruitment of 0.17 offspring per soft-released pair, compared to a local first-year breeding recruitment of 0.05 offspring per hard-released pair. For traditional hard-releases between 2002 and 2004, recruitment of offspring (2003–2005) was similarly low, at 0.06 recruits per pair. Thus, when captive-reared adults are soft-released rather than hard-released, it appears that they can be expected to contribute approximately three times the quantity of offspring to the future wild population; however, several more years of releases, with a large number of released pairs, would be necessary to demonstrate this with statistical significance.

5. Conclusions

Soft-releases of burrowing owls in BC were more successful than hard-releases with respect to each of three short-term objectives:

soft-releases resulted in (1) higher settlement of animals in release areas, (2) higher survival of individuals following release, and (3) higher reproductive output in the wild. Though longer-term success has yet to be determined, our results suggest that soft-released pairs might be as much as three times as successful at contributing first-year recruits into the local breeding population. Given the considerable potential benefits listed above, the placement of release-enclosures for a few weeks at the beginning of each season is a relatively low investment, compared to the increased time, effort, and cost required to produce additional owls in captivity. We therefore recommend that soft-releases be used as the primary reintroduction strategy for future burrowing owl releases in BC and elsewhere in North America.

Based on our success relative to other burrowing owl releases that used a variety of techniques, we recommend that other soft-releases use the same protocol that we did in this study. Specifically, release yearling adult owls, with one male and one female per burrow, at the beginning of each breeding season, using source birds reared in captivity near release sites. Our results, in combination with those of Poulin et al. (2006), indicate that 14–17 days appears to be an appropriate period for pre-release confinement. In addition, we suggest supplemental feeding, throughout the breeding season, to maximize reproductive output.

Our experiment contributes to the growing body of evidence suggesting that soft-released animals often outperform hard-released animals. We suspect the benefits of soft-release may be particularly evident for species with a high tendency for dispersal or those likely to experience significant predation pressure soon after release. We therefore join several previous authors in suggesting that soft-releases be used as the method of choice in reintroductions and translocations for a wide-variety of species (e.g., Bright and Morris, 1994; Wanless et al., 2002; Teixeira et al., 2007).

Lastly, we encourage future controlled experiments to test additional factors that influence the short- and long-term success of species reintroduction programs.

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